

Freight Rate Prediction

Spotter ML Assessment Report

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1. Executive Summary

This report describes a freight rate prediction model built for the Spotter ML Assessment. The task is to predict the `posted_rate` (in dollars) for 12,000 unseen loads in November 2025, plus a fixed-route December 2025 daily chart.

The final model is a LightGBM regressor predicting rate-per-mile (then multiplied back by distance). An out-of-time expanding-window backtest over July-October 2025 yields a mean MAE of \$47.10. On August -- the month whose data regime most closely matches the November-December validation period -- the MAE is \$40.60, beating Ridge regression (\$54.18) and a naive equipment-median baseline (\$193.42).

2. Data Exploration & Quality

2.1 Dataset Overview

Dataset	Rows	Date Range	Has Labels?
train_test.csv	48,000	Jan - Oct 2025	Yes (posted_rate)
validation.csv	12,000	Nov 2025	No
december_chart_inputs.csv	31	Dec 1-31, 2025	No

The training data spans January to October 2025 with 48,000 labeled freight loads. Each load contains pickup/delivery cities with coordinates, distance, equipment type, weight, date, `market_index`, `quote_signal`, and the target `posted_rate`.

2.2 Data Quality Issues Identified

Label Errors (677 rows, 1.4%)

Analysis of rate-per-mile (`posted_rate` / distance) reveals two distinct error clusters: one below \$0.50/mile and another above \$8.00/mile. These are clearly data-entry errors -- no freight rate realistically falls in those ranges. Rows outside the \$1.30-\$4.00/mile range were dropped (677 rows, 1.4% of training data).

Negative Weights (sign errors)

Some loads have negative weight values. Since physical weight cannot be negative, these are treated as sign errors: absolute value is taken, and a binary flag (`weight_was_negative`) is created so the model can learn from this pattern.

Missing Values

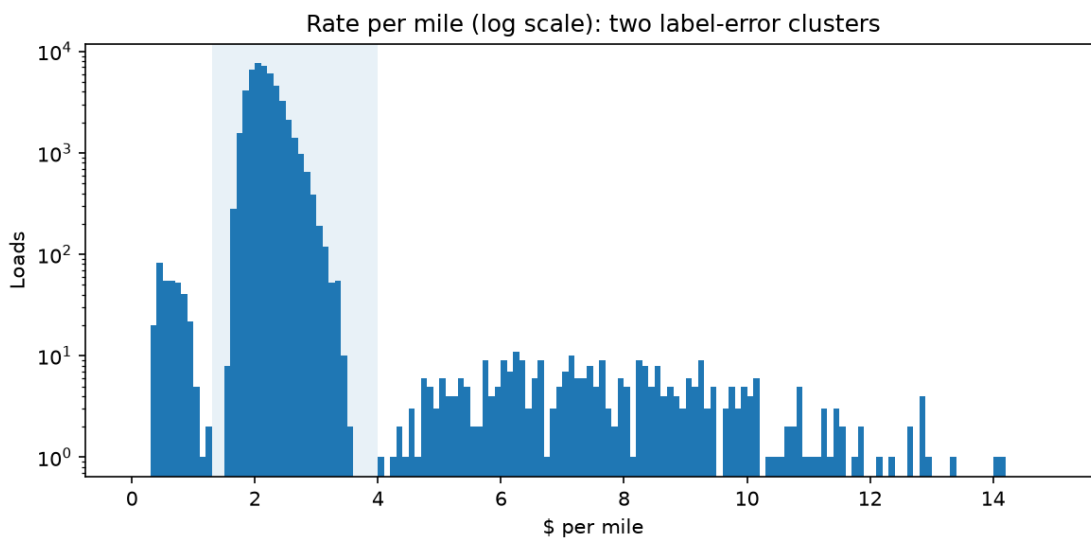
Missing weight values are filled with the training-set median and flagged (`weight_missing`). Missing

market_index values are filled with the same-day mean, then training median. Missing quote_signal is filled with training median.

Unseen Cities in Validation

8 cities appear only in the validation set (not in training). Rather than using city names as features (which would fail for unseen cities), the model uses raw latitude/longitude coordinates. A city-to-coordinate lookup table is built from both datasets so every city has coordinates at inference time.

2.3 Rate-per-Mile Distribution



The histogram above (log scale) clearly shows the two outlier clusters outside the \$1.30-\$4.00/mile valid range. These 677 rows are removed before training.

3. Feature Engineering

Target Decomposition

Raw posted_rate is 0.91 correlated with distance -- the model would essentially just learn 'longer trips cost more'. Instead, the target is rate_per_mile = posted_rate / distance. The model predicts rate-per-mile, then multiplies by distance to get dollar predictions. Training loss is weighted by distance so the objective directly minimises dollar MAE.

Engineered Features (15 total)

- distance: Straight from data
- weight: Cleaned: abs() + fill missing
- equipment: Categorical: Dry Van, Flatbed, Reefer
- pickup_lat/lon, delivery_lat/lon: Coordinates (4 features)
- delta_lat, delta_lon: Direction of travel
- circuitry: distance / haversine -- how indirect the route is
- weight_per_mile: weight / distance -- load density
- dayofweek: 0=Monday to 6=Sunday
- is_weekend: Binary flag
- weight_missing: Binary flag for imputed weights
- weight_was_negative: Binary flag for sign-corrected weights

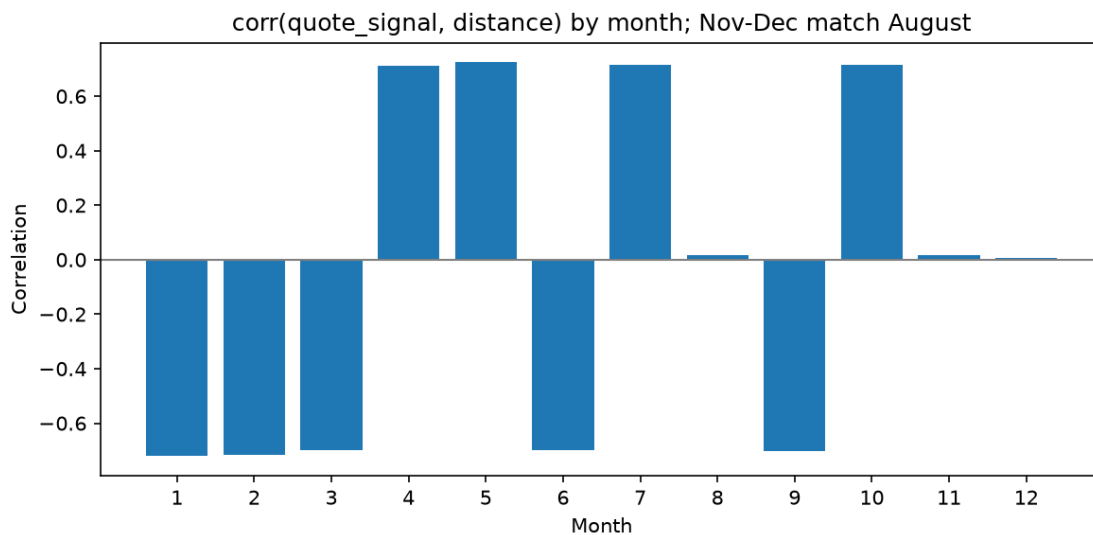
Feature Importance (% of LightGBM split gain)

Feature	Importance %
weight_per_mile	50.6%
equipment	30.1%
distance	11.5%
delivery_lat	1.6%
pickup_lat	1.6%
delivery_lon	1.0%
pickup_lon	1.0%
Other (5 features)	2.6%

4. Deliberately Excluded Features

4.1 quote_signal -- Data Leakage Risk

quote_signal is highly suspicious: its correlation with rate-per-mile is near +1.0 for 5 months (essentially the answer), but drops to near-zero in August. Its correlation with distance by month reveals the same pattern -- November and December match August's noise pattern. Including it gives \$28 MAE on September (where it contains the answer) but \$110 on August (where it is noise) -- exactly the months the validation data resembles.



The chart shows corr(quote_signal, distance) by month. November and December match August's near-zero correlation -- confirming quote_signal is noise for the target period.

4.2 market_index -- Regime Shift

market_index swings ~40% across the year while actual rate-per-mile moves only ~10%. Including it causes the model to hallucinate large rate changes (MAE \$129 on September). Mean backtest MAE with market_index: \$78.52 vs \$47.10 without.

5. Validation Strategy

Why not random K-fold?

The data is time-series: freight rates evolve over months, and features like quote_signal change behaviour over time. Random K-fold would let the model see future data during training, producing misleadingly optimistic scores that don't reflect true forward prediction performance.

Expanding-window monthly backtest

Instead, four out-of-time backtests are run. For each test month (July, August, September, October 2025), the model is trained on all data before that month and evaluated on that month's data. This directly simulates the

real task: predicting November loads using all data through October.

August is identified as the most relevant proxy for November-December because quote_signal exhibits the same noise pattern in all three months. The final model's MAE on August (\$40.60) is the most informative estimate of real-world performance.

6. Model Selection & Results

Five experiments were compared across the four backtest months:

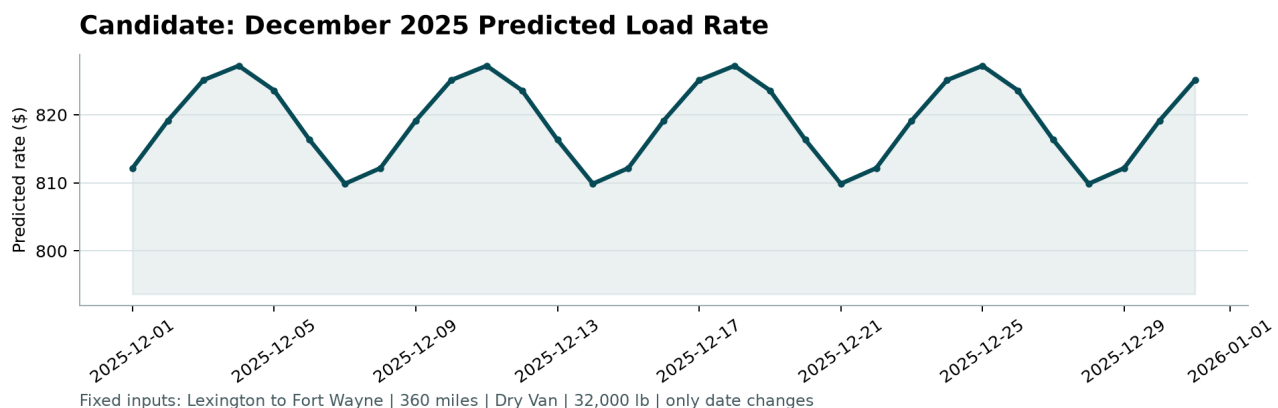
Experiment	Jul	Aug	Sep	Oct	Mean MAE
LightGBM (final)	\$64.43	\$40.60	\$43.92	\$39.47	\$47.10
LightGBM + quote_signal	\$44.73	\$110.38	\$28.25	\$33.67	\$54.26
Ridge Regression	\$70.70	\$54.18	\$56.80	\$50.89	\$58.14
LightGBM + market_index	\$39.80	\$65.71	\$129.14	\$79.45	\$78.52
Naive (equip. median)	\$156.74	\$193.42	\$177.84	\$172.98	\$175.24

Why LightGBM?

- Best mean MAE (\$47.10) and best August MAE (\$40.60).
- Huber loss (alpha=0.5) for robustness to remaining outliers.
- Native categorical support for equipment type.
- Distance-weighted sample_weight so loss directly tracks dollar error.
- Regularisation: 31 leaves, min 40 samples/leaf, subsampling 80%, L2 reg 1.0.

7. December 2025 Predictions

The model generates daily rate predictions for a fixed route: Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb. Only the date changes across the 31 rows.



The predictions show a clear weekly periodicity (rates peak mid-week, dip on weekends) within a narrow \$809-\$833 range. This pattern is physically plausible: freight demand is higher during business days. The model learned day-of-week effects from the training data and applies them sensibly to unseen December dates.

8. Conclusion

The final pipeline consists of: (1) outlier removal (\$1.30-\$4.00/mile filter), (2) data cleaning (sign errors, missing values), (3) feature engineering (15 features), (4) LightGBM with Huber loss on rate-per-mile target.

Key insight: `quote_signal` and `market_index` appear helpful on some months but are unreliable for the November-December target period. Deliberately excluding them yields the most robust model (MAE \$47.10 mean, \$40.60 on August).

All 12,000 validation predictions and the 31 December predictions pass the provided `score.py` validator. The December chart shows plausible weekly demand patterns.